

SHAHZAD ALI

PhD Researcher · Graph Learning & AI for Neurodegenerative Disease Modelling

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PROFESSIONAL SUMMARY

PhD Researcher in Data Science and Computation at Alma Mater Studiorum – Università di Bologna, based at the IRCCS Ospedale Policlinico San Martino (LISCOMP Lab), Genova. Research focus on graph learning and AI methods for neurodegenerative disease modelling, with emphasis on Graph Neural Networks and multimodal neuroimaging. 10 peer-reviewed journal articles, 5 conference papers, and 3 conference abstracts; 279 citations, h-index 6, i10-index 2 (Google Scholar, 2026). Combines strong machine and deep learning skills with hands-on neuroimaging pipelines (FreeSurfer, MRtrix3, FSL, SPM, Clinica, Nilearn) for clinically relevant AI applications.

EDUCATION

PhD in Data Science and Computation, Alma Mater Studiorum – Università di Bologna *Bologna, Italy · Nov 2022 – Present*

Field: Machine Learning Methods in Digital Health · 38th cycle, funded by Next Generation EU (NRRP, DM 351/2022).

Thesis (in progress): Graph Learning and AI Methods for Neurodegenerative Disease Modelling.

Research carried out at the LISCOMP Laboratory, IRCCS Ospedale Policlinico San Martino, Genova.

M.Sc. in Computer Science, University of Engineering and Technology (UET), Lahore *Lahore, Pakistan · Jan 2016 – Jul 2018*

Specialisation: Machine Learning · GPA: 3.75 / 4.00.

Thesis: Artificial Neural Network Model for Rainfall-Runoff Prediction for Jhelum River Basin Upstream of the Mangla Dam, Pakistan.

Awarded the Punjab Educational Endowment Fund (PEEF) Talent-Based Scholarship.

B.Sc. in Telecommunication and Networking, COMSATS University Islamabad, Sahiwal Campus *Sahiwal, Pakistan · Aug 2011 – Jun 2015*

Fields: Computer Science, Telecommunication and Networking · GPA: 3.46 / 4.00.

Thesis: Visible Light-Based Communication System for High Data Rate Real-Time Applications.

Final Year Project funded by the ICT R&D Programme; ICT R&D Merit Scholarship throughout the degree.

RESEARCH & WORK EXPERIENCE

PhD Researcher, IRCCS Ospedale Policlinico San Martino *Genova, Italy · Sep 2023 – Present*

- Applied machine learning, deep learning, and Graph Neural Networks for disease diagnosis using multimodal neuroimaging data (sMRI, dMRI, fMRI, amyloid-PET).
- Developed AI models to enhance diagnostic accuracy for neurological disorders, with focus on Alzheimer's disease staging and cognitive decline prediction.
- Built end-to-end neuroimaging pipelines (FreeSurfer, MRtrix3, FSL, SPM, Clinica, Nilearn) on large clinical cohorts.
- Collaborated with clinicians, radiologists, and statisticians at the LISCOMP Lab; published 10 peer-reviewed journal articles during this position.

Visiting Researcher, LILI Lab — Learners of Interpretable Latent Information, University of Sussex Brighton, *United Kingdom · Jul 2025 – Jan 2026*

- Research on AI and graph learning methods for neurodegenerative disease modelling; emphasis on clinically relevant and translational applications.
- Designed interpretable graph-based architectures for structured clinical and healthcare data; strengthened international collaboration between LISCOMP / MIDA groups and the UK research community.

Lecturer in Computer Science, University of Education, Lahore

Lahore, Pakistan · Mar 2019 – Present

- Teaching undergraduate-level courses in Data Mining, Artificial Intelligence, and Introduction to Machine Learning.
- Supervising final-year undergraduate projects across AI, data analytics, and software systems.
- Attending and speaking at conferences and seminars; representing the department at national and international venues.

Graduate Research Assistant, University of Engineering and Technology (UET), Lahore *Lahore, Pakistan · Oct 2016 – May 2018*

- Designed and taught the laboratory subject of Artificial Intelligence (Fall 2016, Fall 2017).
- Assisted faculty in coursework delivery and final-year project supervision alongside MSc research.

TECHNICAL SKILLS

Programming: Python (Expert), C++ (Expert), R, Java

Machine & Deep Learning: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, XGBoost, GradCAM

Graph & Network ML: Graph Neural Networks, PyTorch Geometric, NetworkX, DGL

Neuroimaging: FreeSurfer, MRtrix3, FSL, SPM, Clinica, Nilearn, DICOM/NIfTI workflows

Data Science: Jupyter, Google Colab, Pandas, NumPy, Matplotlib, Seaborn

ML Platforms: Azure ML Studio, Dataiku, RapidMiner, Weka, KNIME

Visualisation: Power BI, Tableau, Plotly

Languages: Urdu (native), English (C2 understanding, C1 speaking & writing)

SELECTED PUBLICATIONS

10 peer-reviewed journal articles · 5 conference papers · 3 conference abstracts. 279 citations · h-index 6 · i10-index 2 (Google Scholar). Full list: <https://alishahzi.github.io/>

Journal Articles (first / co-author, selected):

- Argenti L, Cirone A, Massa F, ..., Ali S, ..., Pardini M (2026). Exploring the spatial covariance of cerebral vascular density and amyloid burden in Alzheimer's disease. *Alzheimer's & Dementia: DADM*, 18(2): e70390. doi:10.1002/dad2.70390
- Ali S, Kreshpa W, Rosso N, Piana M, Roccatagliata L, Cirone A, Lorenzini L, Campi C, Pardini M, Garbarino S (2026). A systematic study on the integration of MRI connectivity metrics for Alzheimer's diagnosis, staging, and cognitive decline prediction. *Frontiers in Neuroimaging*, 5:1746464. doi:10.3389/fnimg.2026.1746464
- Argenti L, Massa F, Losa M, ..., Ali S, Piana M, ..., Pardini M (2026). Neurotransmitter landscape and neurodegeneration patterns in Alzheimer's Disease. *Neurobiology of Aging*, 162: 49-56. doi:10.1016/j.neurobiolaging.2026.02.004
- Losa M, Cotta Ramusino M, ..., Ali S, ..., Garbarino S, Roccatagliata L, Farina LM, Pardini M (2025). Cerebrospinal Fluid Biomarkers Profiling in Cerebral Amyloid Angiopathy and Relationship With Disease Phenotypes. *J. American Heart Association*, 14(20): e044784. doi:10.1161/JAHA.125.044784
- Ali S, Piana M, Pardini M, Garbarino S (2025). Graph neural networks in Alzheimer's disease diagnosis: a review of unimodal and multimodal advances. *Frontiers in Neuroscience*, 19: 1623141. doi:10.3389/fnins.2025.1623141
- Kiran I, Ali S, Alhussein M, Aslam S, Aurangzeb K (2025). An AI-Enabled Framework for Transparency and Interpretability in Cardiovascular Disease Risk Prediction. *Computers, Materials & Continua*, 82(3): 5057-5078. doi:10.32604/cmc.2025.058724
- Ali S, Shahbaz M (2020). Streamflow forecasting by modeling the rainfall–streamflow relationship using artificial neural networks. *Modeling Earth Systems and Environment*, 6(3): 1645-1656. doi:10.1007/s40808-020-00780-3

Conference Abstracts & Posters (Presented, 2024–2026):

- Ali S et al. (2026). MRI Connectivity Metrics for Alzheimer's Diagnosis, Staging, and Cognitive Decline Prediction — Organization for Human Brain Mapping (OHBM) Annual Meeting, Bordeaux, France.
- Ali S et al. (2026). Hierarchical Graph-of-Graphs Learning for Multimodal Alzheimer's Disease Staging and Progression Modelling — Learning on Graphs Italian Meetup 2026, Pisa, Italy.

- Ali S, Kreshpa W, Piana M, Pardini M, Garbarino S (2024). Integration of Graph Theory and Brain Microstructural Analysis with Machine Learning for Advanced Diagnosis of Alzheimer's Disease and Prediction of Cognitive Decline — IV Annual Meeting RIN IRCCS Network — Symposia, Italy.

AWARDS & RECOGNITION

- PhD Scholarship — Alma Mater Studiorum Università di Bologna · Next Generation EU (NRRP, DM 351/2022) (2022).
- Punjab Educational Endowment Fund (PEEF) Talent-Based Scholarship — Government of Punjab, Pakistan (2016).
- ICT R&D Final Year Project Grant — ICT R&D Programme, Federal Government of Pakistan (2014).
- ICT R&D Merit Scholarship (BSc) — Federal Government of Pakistan (2011).

PEER REVIEW

Frontiers in Neuroscience · Frontiers in Neuroimaging · Computers, Materials & Continua · IEEE Access · Modeling Earth Systems and Environment (Springer).

CONTINUING PROFESSIONAL DEVELOPMENT

Coursera / IBM micro-credentials in Computational Neuroscience, Deep Learning (PyTorch · GradCAM), Generative AI (Prompt Engineering · Applications), Machine Learning Pipelines (Azure ML Studio · Dataiku), and Data Visualization — 12+ courses (2024–2025).

Schools & summer programmes: Geilo Winter School on Graphs and Applications (SINTEF, Norway, 2024); Eastern European Machine Learning (EEML) Summer School (Košice, Slovakia, 2023); IEEE ComSoc eHealth TC PhD School (Genova, 2023); EURO-PAD Scientific Symposium (Genova, 2026); Invited speaker — Data Science and Machine Learning for Physicists (NCP, Islamabad, 2019).